Report 1: Shgian, Delima (File Management)

Individual Contribution Report

- **Student Name:** Shgian [Insert Last Name]
- **Group:** SureballNa (MiniOSNova)
- **Assigned Role:** File Management Developer

What did I contribute to the project?

- Developed the virtual file system module (filesystem.py) to handle in-memory file and folder operations.
- Assisted in testing file commands in the terminal and writing the file management section of the User Manual.

What features did I personally develop?

- I help programmed the folder and file creation functions (mkdir, create, and dir) using Python data structures to store virtual items in memory.
- Implemented the read and delete functions to view file contents and safely remove files from the virtual directory.

What problems did I encounter?

- Creating a file or folder with a name that already existed overwrote the old one without warning the user.
- Trying to read or delete a file that did not exist caused a Python KeyError during initial testing.

How did I solve those problems?

- Added a conditional check (if name in) before creating a file or folder so the system warns the user if the name is already taken.
- Added a validation check before running read or delete so the program prints a friendly "File not found" message instead of crashing.

What did I learn from this project?

I learned how an operating system simulates a virtual directory structure in memory using Python dictionaries and lists, and how separate module files interact with a main program.

Evidence of my contribution:

- **Source Code:** Implementation of filesystem.py in the official repository (https://github.com/markelleryc-cpu/MiniOSNova).

- **Project Documentation:** File management command tests and contributions to the project documentation in the docs/ folder.